

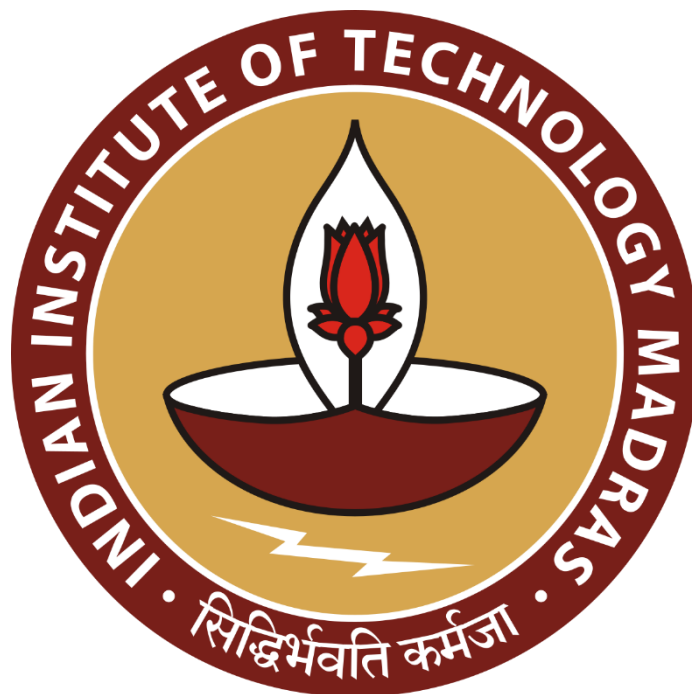
Optimizing Customer Engagement and Inventory Management at a Small Supermarket

A Proposal report for the BDM capstone Project

Submitted by

Name: Wamiq Mushtaq

Roll number: 22f1001807



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project titled “**Optimizing Customer Engagement and Inventory Management at a Supermarket**”.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Wamiq Mushtaq

Signature of Candidate: **(Digital Signature)**

Name: Wamiq Mushtaq

Date: 07/03/2025

1.0 Executive Summary and Title

This BDM capstone project's aim is to aid a supermarket in Malaysia, City Mall Holdings, in enhancing its operational efficiency and profitability. Addressing poorly understood consumer engagement stemming from unfocused marketing, along with ineffective inventory management that results in stockouts and overstocking is the key to this project. I will apply data analytics techniques to the supermarket's sales data in order to gain knowledge about the customer's purchasing behavior. Through the application of various segmentation techniques such as RFM (recency, frequency, monetary), demographic, and payment method, marketing strategies that cater to the particular needs of every segment will be designed. This will increase sales revenue. Furthermore, I'll apply time series forecasting to stock analysis in order to increase the profit margin, while reducing expenses and enhancing customer satisfaction. Greater understanding of customer needs is the profound expectation of this study.

2.1 Organization Background

I'm utilizing a supermarket dataset for my project, which is accessible on Kaggle, a well-known website that hosts datasets of all types. The Dataset belongs to one of the leading retail outlets in Myanmar called City Mall Holdings. City Mall Holdings has grown to become "the grocery market leader in Myanmar and an icon in the country's retail industry".

They operate several branches around Myanmar in cities like Yangon and Mandalay. It is a business-to-consumer company that specializes on technology, food, cosmetics, health, and home and leisure products. Since customers are essential to business operations, it would be beneficial to learn their preferences as the supermarket is positioned as a local provider.

3.1 Problem Statement

1.

The following goals are intended to be accomplished by this initiative for the Supermarket:

3.1 Improve Customer Engagement Create focused marketing plans to boost consumer response rates and foster brand loyalty. There is a chance to create campaigns that are aimed at every group.

3.2 Optimize Levels of Inventory: Use data-driven inventory management techniques to enhance product availability, limit overstocking, and decrease stockouts. We may decrease inventory stockouts by using forecasting algorithms, which raises customer satisfaction.

3.3 Increase Profitability: By raising sales, cutting expenses, and enhancing operational effectiveness, you can raise total profitability. By implementing these strategies, we can boost our annual revenue and improve efficiency

4.1 Background of the Problem

A few important takeaways emerged from an initial data study of the Supermarket Dataset:

First of all, although customers are widely divided, there is hardly any variation in promotions amongst client types, which led me to conclude that they could increase customer engagement. Second, an examination of important product categories including home/lifestyle and health/beauty showed many days with numerous stockouts, highlighting the inefficiencies of the inventory.

This was clear from the negative slope of ratings during those periods, which indicated that repeated stockouts had resulted in low consumer satisfaction. Furthermore, information revealed unused payment methods that can reduce transaction costs.

These deficiencies will be filled by the project, which will also give the supermarket suggestions and resources to help them compete more successfully. Internal and external issues are the main obstacles, and the company will need to change in order to survive. In order to survive in the market, the company must investigate customer-specific insights as much as feasible.

5.1 Problem Solving Approach

I will use a variety of methods in order to gain useful information and apply relevant practical implementations, which include: This project will tackle customer engagement issues along with inefficient inventory management using a systematic, analytical approach.

Data-Driven Customer Segmentation: In this approach, I will analyze customer demographics such as gender and customer type, as well as their purchase frequency and spending patterns. I'll begin with performing Exploratory Data Analysis (EDA) and some basic statistical analysis like calculating averages and distributions. After identifying preliminary patterns, I will implement a more sophisticated technique known as clustering. By employing K-Means Clustering, an unsupervised machine learning algorithm, customers will be segmented automatically according to their purchasing habits. This will enable a more optimal classification of the customer base to increase efficiency, reach, and returns.

Promotional focus: For each target market, I will craft focused plans by understanding what services and products appeal to them.

Inventory Analysis and Optimization through Data Visualization: I will use basic time series analysis and visualizations of sales patterns including trends and stockouts to construct estimates for demand of certain product lines and make appropriate quick adjustments for high and low performing assets.

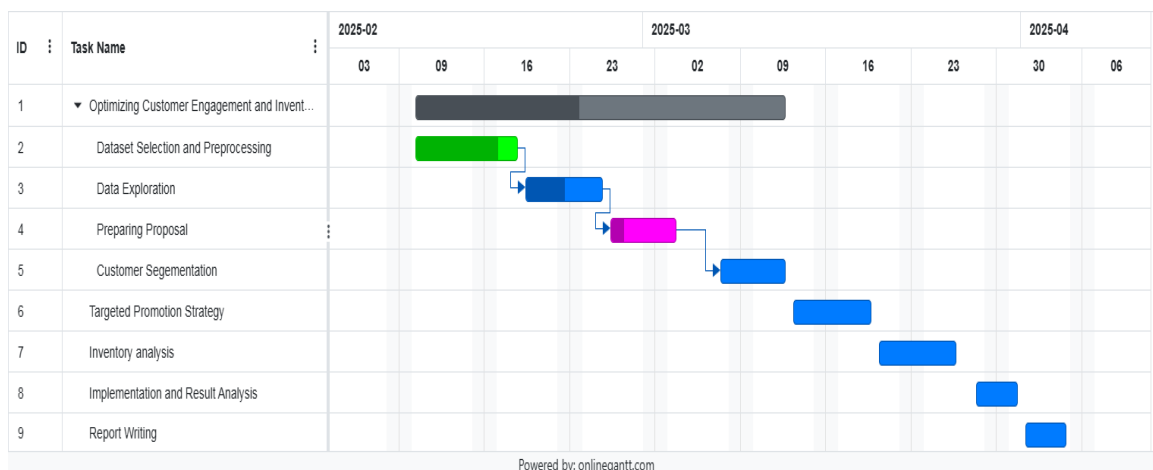
This hybrid approach makes sure that both basic techniques and advanced complex modelling tools are employed to maximize effectiveness of results. As with any data driven process, I have scoped out risk scenarios and planned in advance, together with ensuring that they can be further followed for scalability.

6.1 Expected Timeline

- **February 2025: Data Preparation**
 - Dataset Selection and Acquisition
 - Data Cleaning and Preprocessing
 - Exploratory Data Analysis (EDA)
- **Early March 2025: Proposal Preparation and Submission**
- **Mid-March 2025: Analysis and Strategy Development**
 - Customer Segmentation (RFM, Demographics, Payment)
 - Targeted Promotion Strategy Design

- Inventory Analysis and Demand Forecasting
- **Late March 2025: Implementation and Evaluation**
 - Promotional Strategy Implementation
 - Performance Analysis and Iteration
- **Early April 2025: Report and Presentation**
 - Report Writing (Findings, Recommendations, Limitations)
 - Presentation Preparation

6.2 Gantt chart



7.1 Expected Outcome

The anticipated results of the initiative are intended to give the supermarket a data-driven basis for better decision-making and business improvement. Important results include:

- **Clearly Defined Customer Segments:** Improving customer engagement can be achieved by developing user-friendly customer segments.

- **Optimized Inventory Management:** Product waste can be decreased and customer satisfaction metrics can be raised with the right knowledge.
- **Data-Driven Business Insights:** Making the decision to be data-driven and make changes is made easier with the help of an appropriate dashboard.